

E9 205 Machine Learning for Signal Processing

**ML, MAP, MMSE and Gaussian
Modeling**

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Instructor - Sriram Ganapathy (sriram@ee.iisc.ernet.in)

Teaching Assistant - Prachi Singh (prachisingh@iisc.ac.in).



Decision Theory (PRML Chap. 1.5)

- ❖ Decision Theory
 - ❖ Inference problem
 - ❖ Finding the joint density $p(\mathbf{x}, \mathbf{t})$
 - ❖ Decision problem
 - ❖ Using the inference to make the classification or regression decision

Decision Problem - Classification

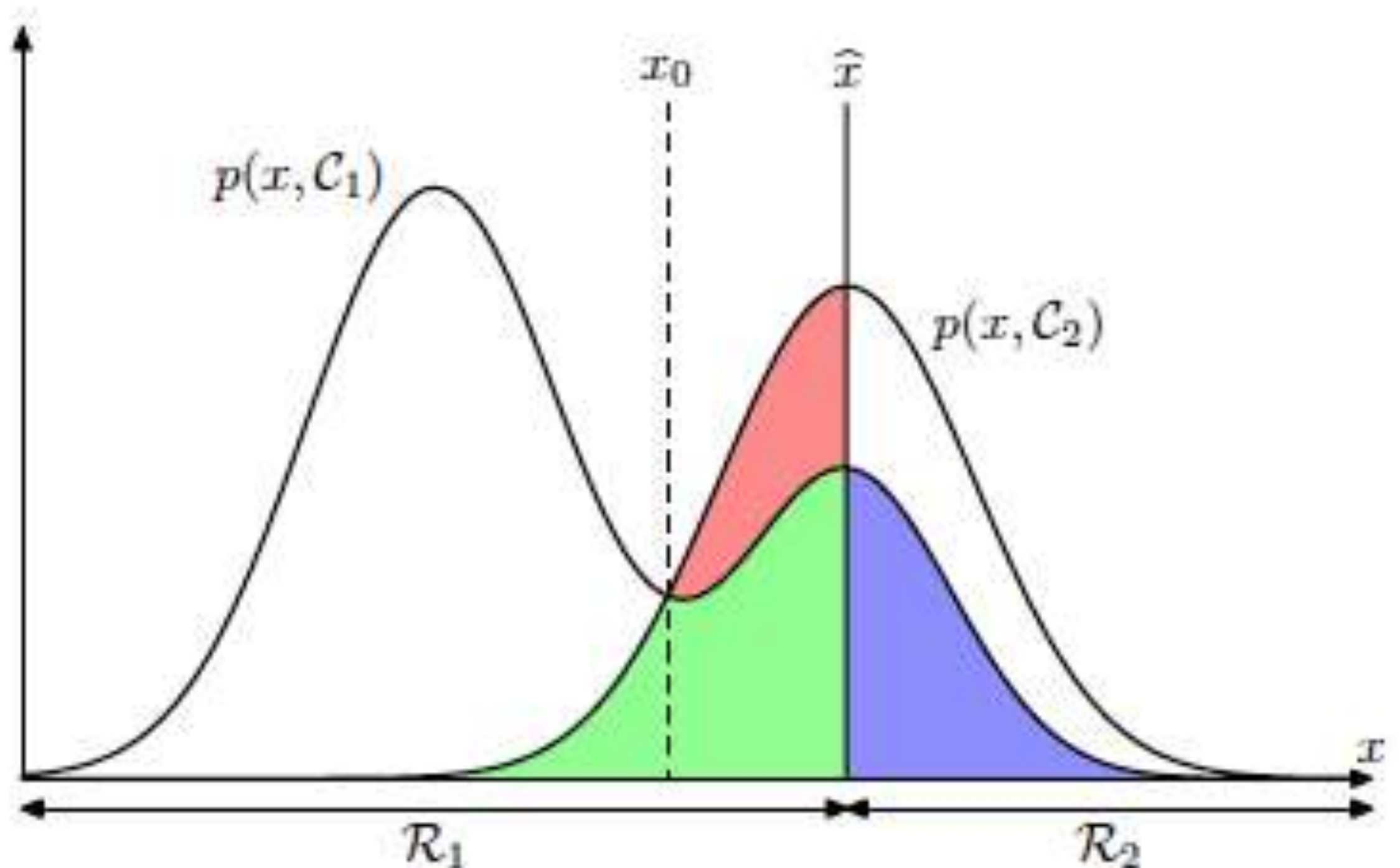
- ❖ Minimizing the mis-classification error
- ❖ Decision based on maximum posteriors

$$\operatorname{argmax}_j p(C_j|\mathbf{x})$$

- ❖ Loss matrix
 - ❖ Minimizing the expected loss

$$\operatorname{argmax}_j \sum_k L_{k,j} p(C_k|\mathbf{x})$$

Visualizing the Max. Posterior Classifier



Approaches for Inference and Decision

I. Finding the joint density from the data.

$$p(C_k|\mathbf{x}) \propto p(\mathbf{x}|C_k)p(C_k)$$

II. Finding the posteriors directly.

III. Using discriminant functions for classification.

Approaches for Inference and Decision

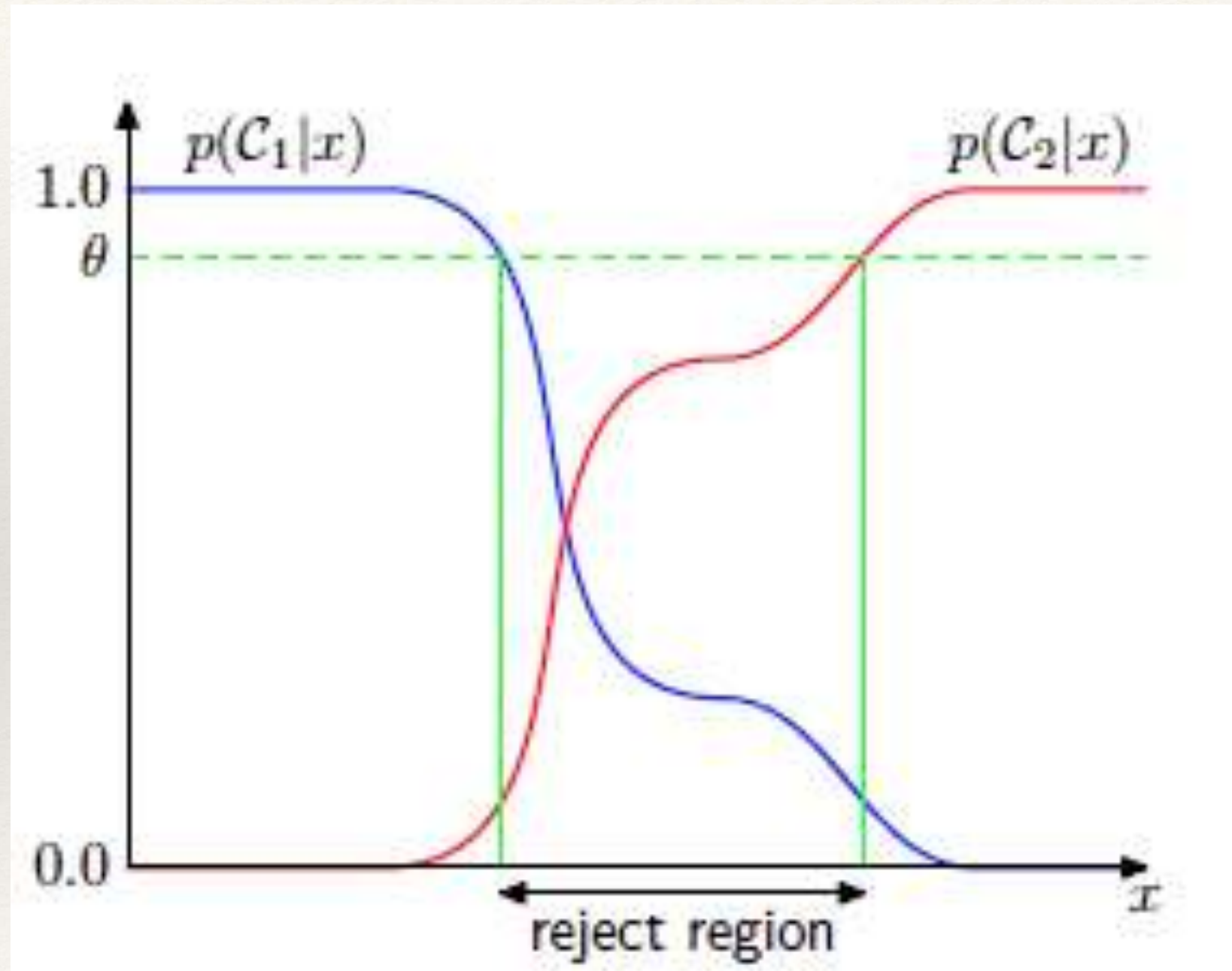
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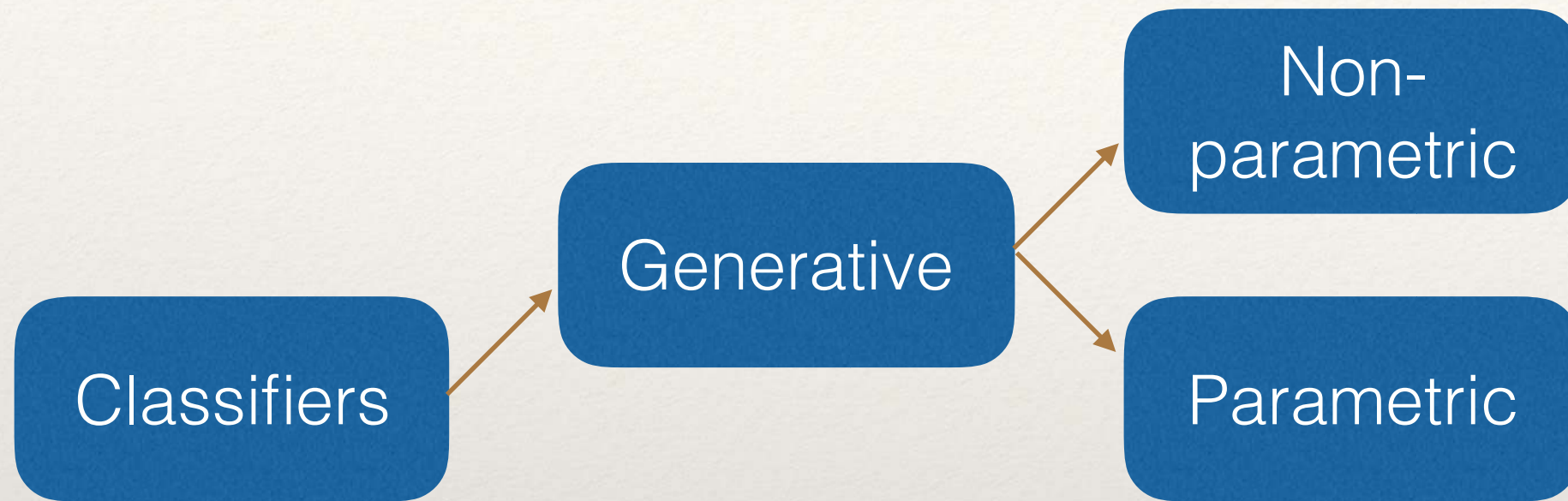
Advantage of Posteriors



Decision Rule for Regression

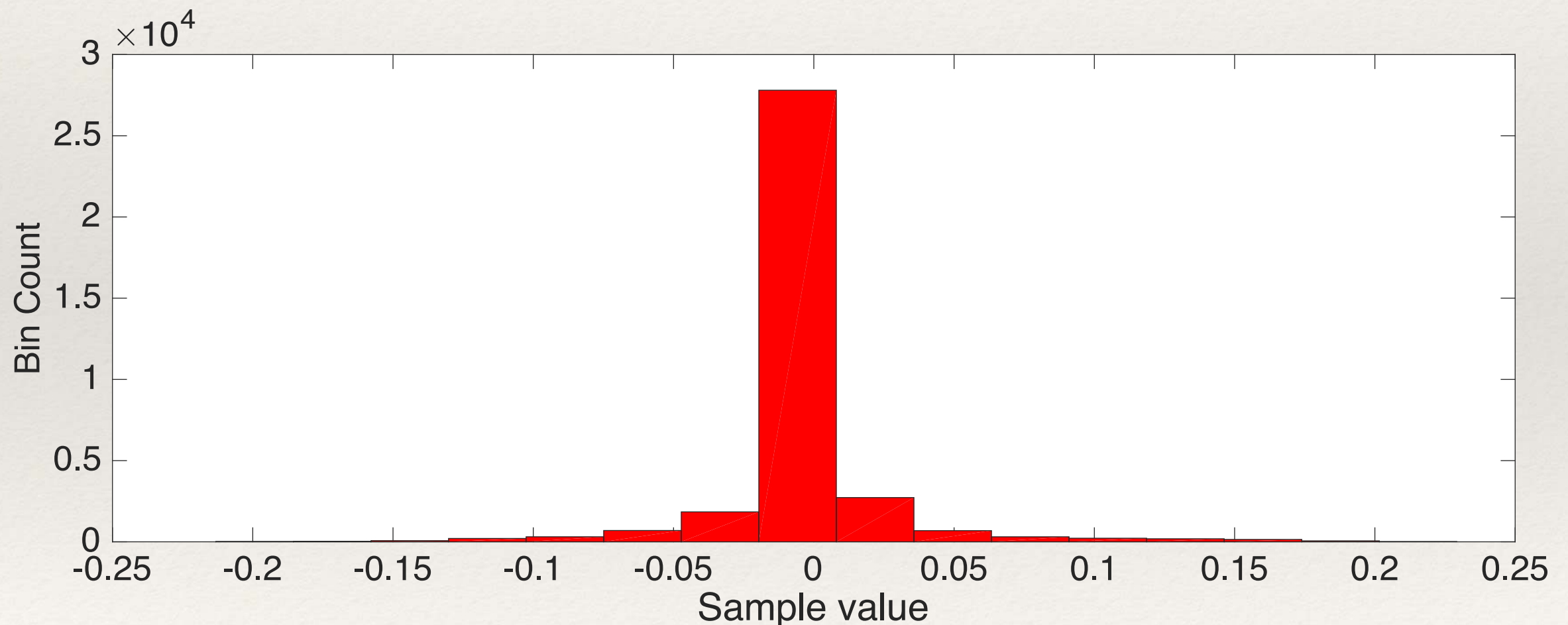
- ❖ Minimum mean square error loss
- ❖ Solution is conditional expectation.

Generative Modeling



Non-parametric Modeling

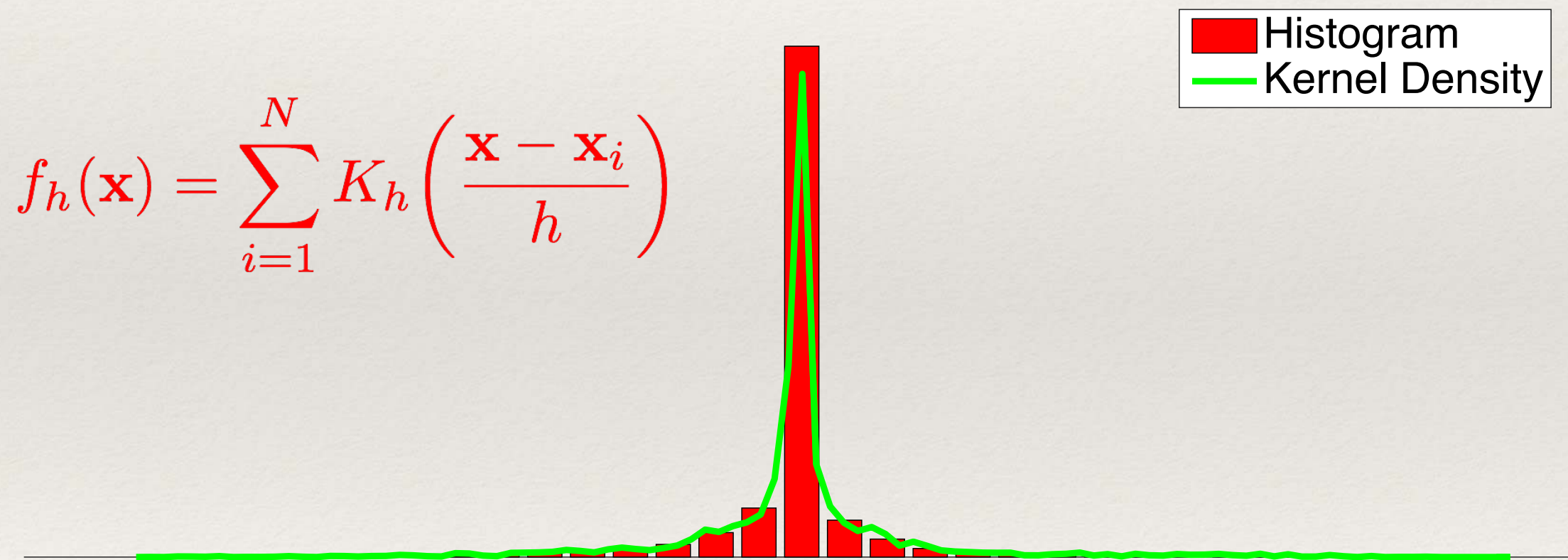
- **Non-parametric** models do not specify an apriori set of parameters to model the distribution. Example - Histogram



The density is not smooth and has block like shape.

Non-parametric Modeling

- **Non-parametric** models do not specify an apriori set of parameters to model the distribution.
 - Example - Kernel Density Estimators



Kernel is a smooth function which obeys certain properties

Non-parametric Modeling

- Non-parametric methods are dependent on number of data points
 - Estimation is difficult for large datasets.
- Likelihood computation and model comparisons are hard.
- Limited use in classifiers

Parametric Models (Chap 2 PRML)

- ❖ Collection of probability distributions which are described by a finite dimensional parameter set

$$\boldsymbol{\theta} = (\theta_1, \theta_2, \dots, \theta_K) \quad P = \{P_{\boldsymbol{\theta}}\}$$

- Examples -

- Poisson Distribution

$$p_{\lambda}(j) = \frac{\lambda^j}{j!} e^{-\lambda}$$

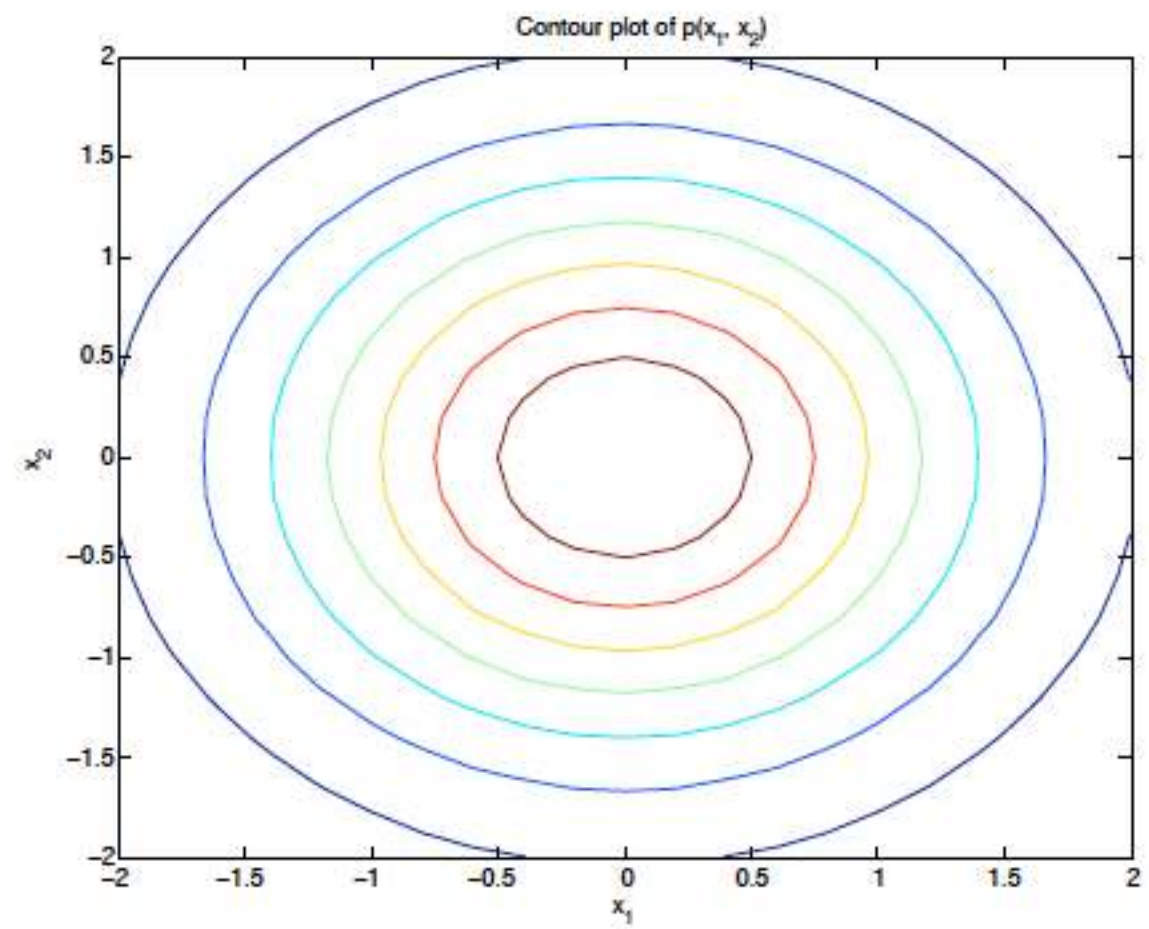
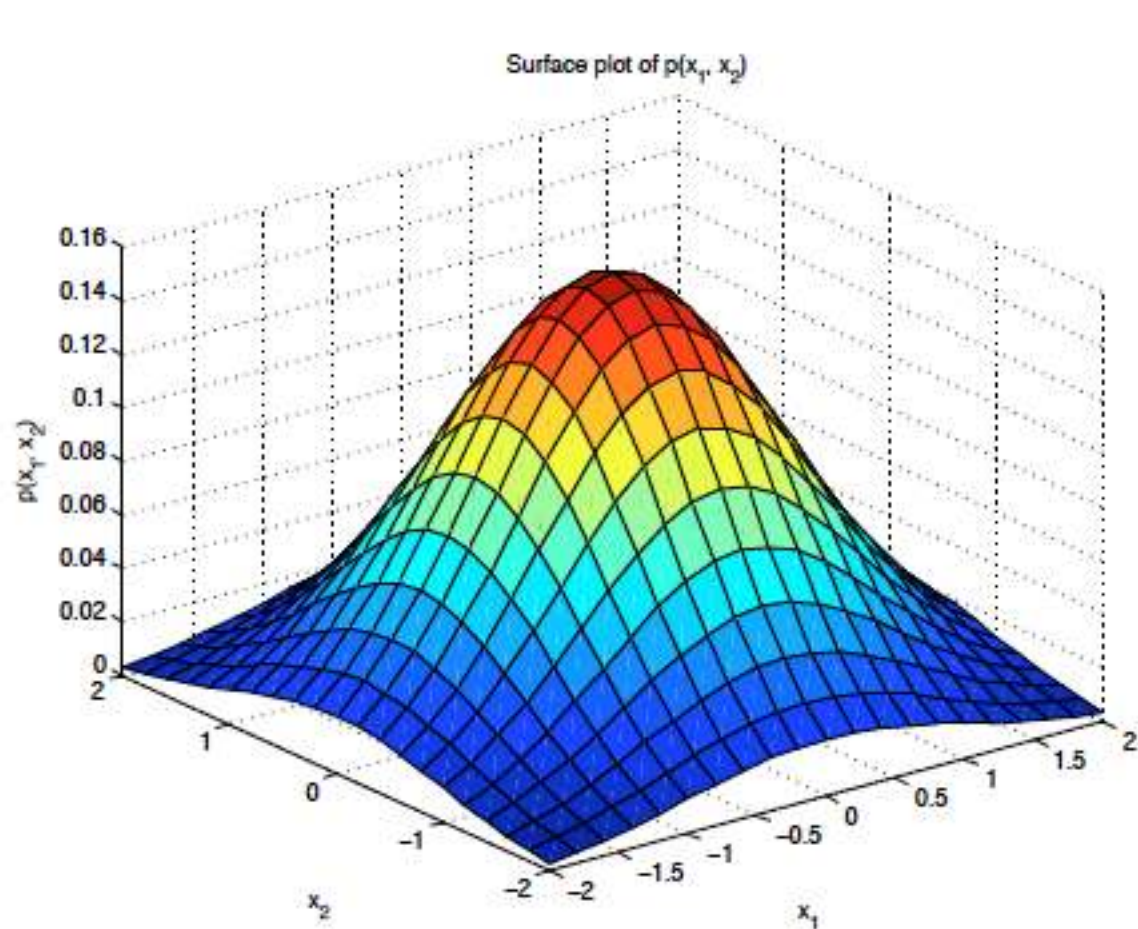
- Bernoulli Distribution

$$p(\mathbf{x}|\boldsymbol{\mu}) = \prod_{i=1}^D \mu_i^{x_i} (1 - \mu_i)^{1-x_i}$$

- Gaussian Distribution

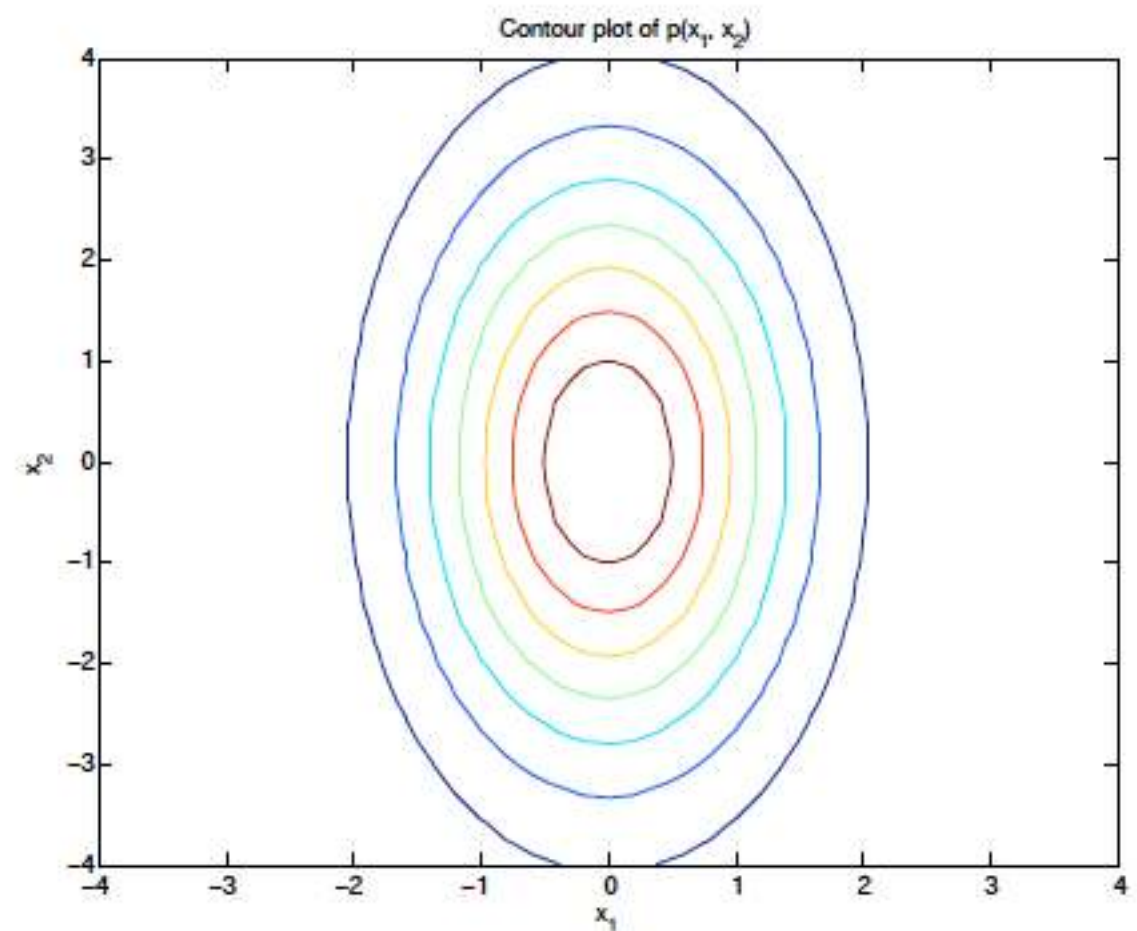
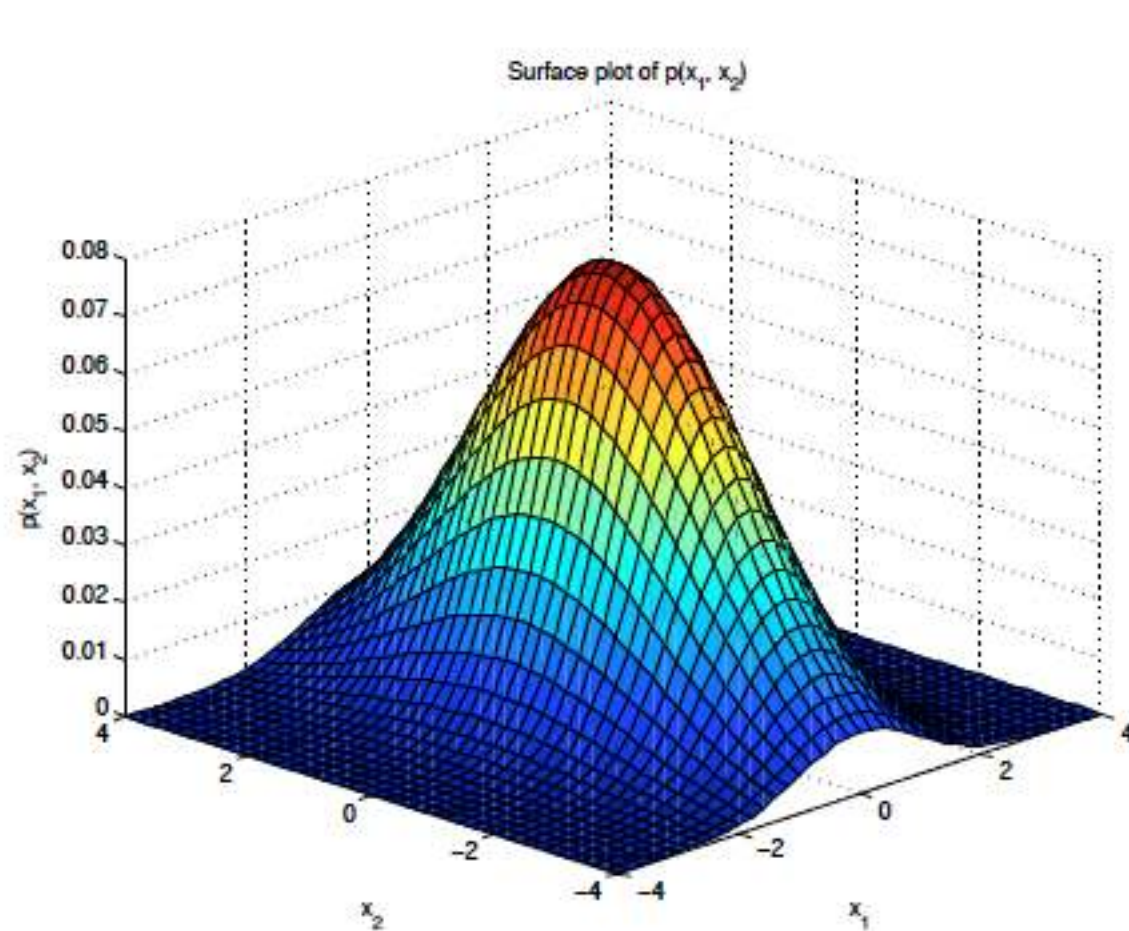
$$p(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{\sqrt{(2\pi)^D |\boldsymbol{\Sigma}|}} \exp \left\{ -\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^* \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right\}$$

Gaussian Distribution



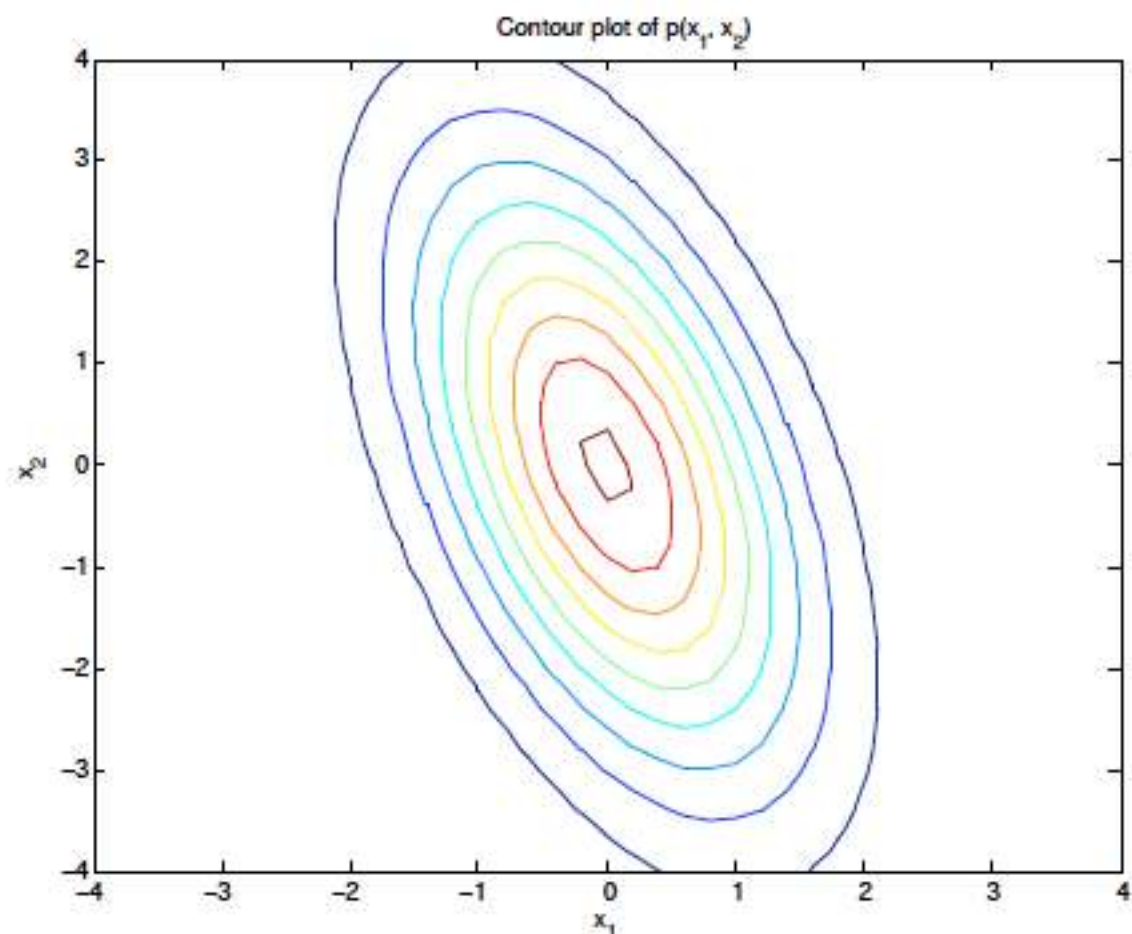
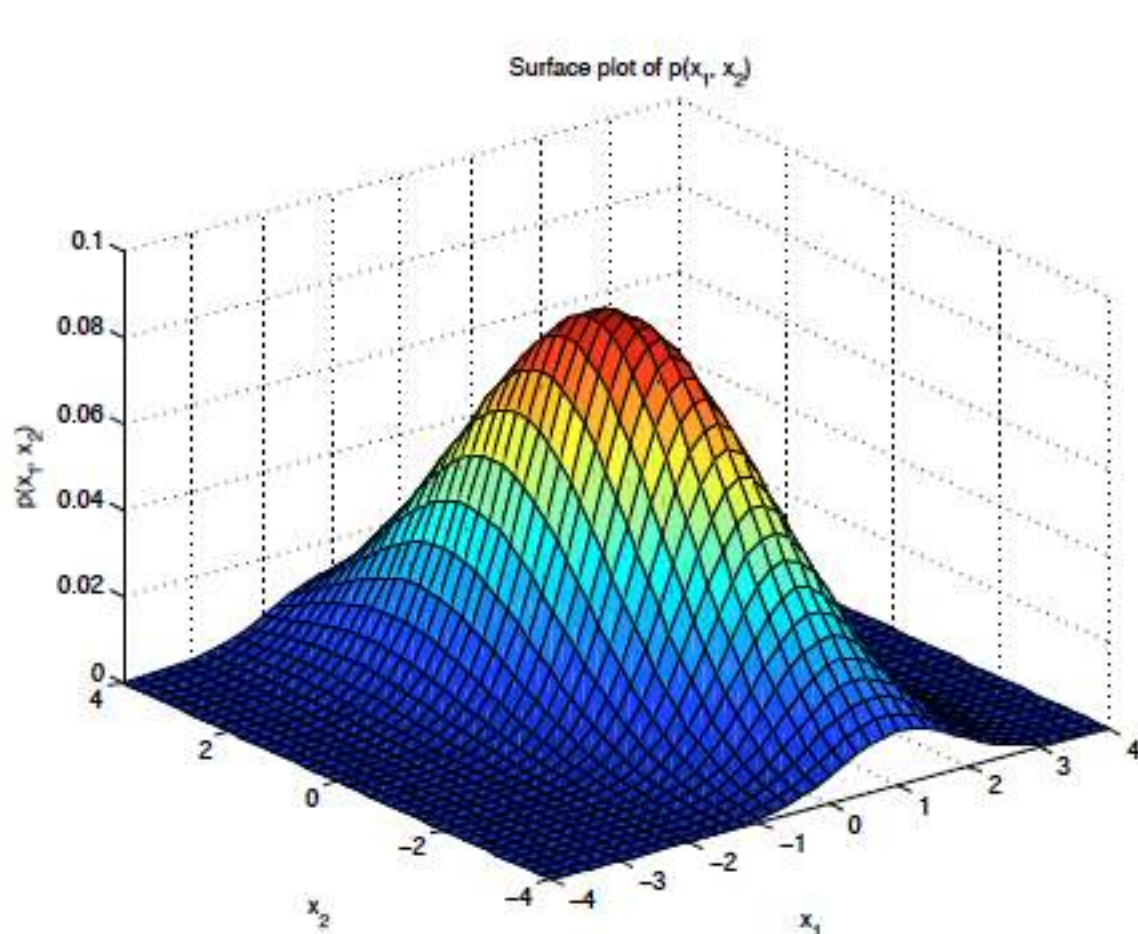
Points of equal probability lie on on contour
Diagonal Gaussian with Identical Variance

Gaussian Distribution



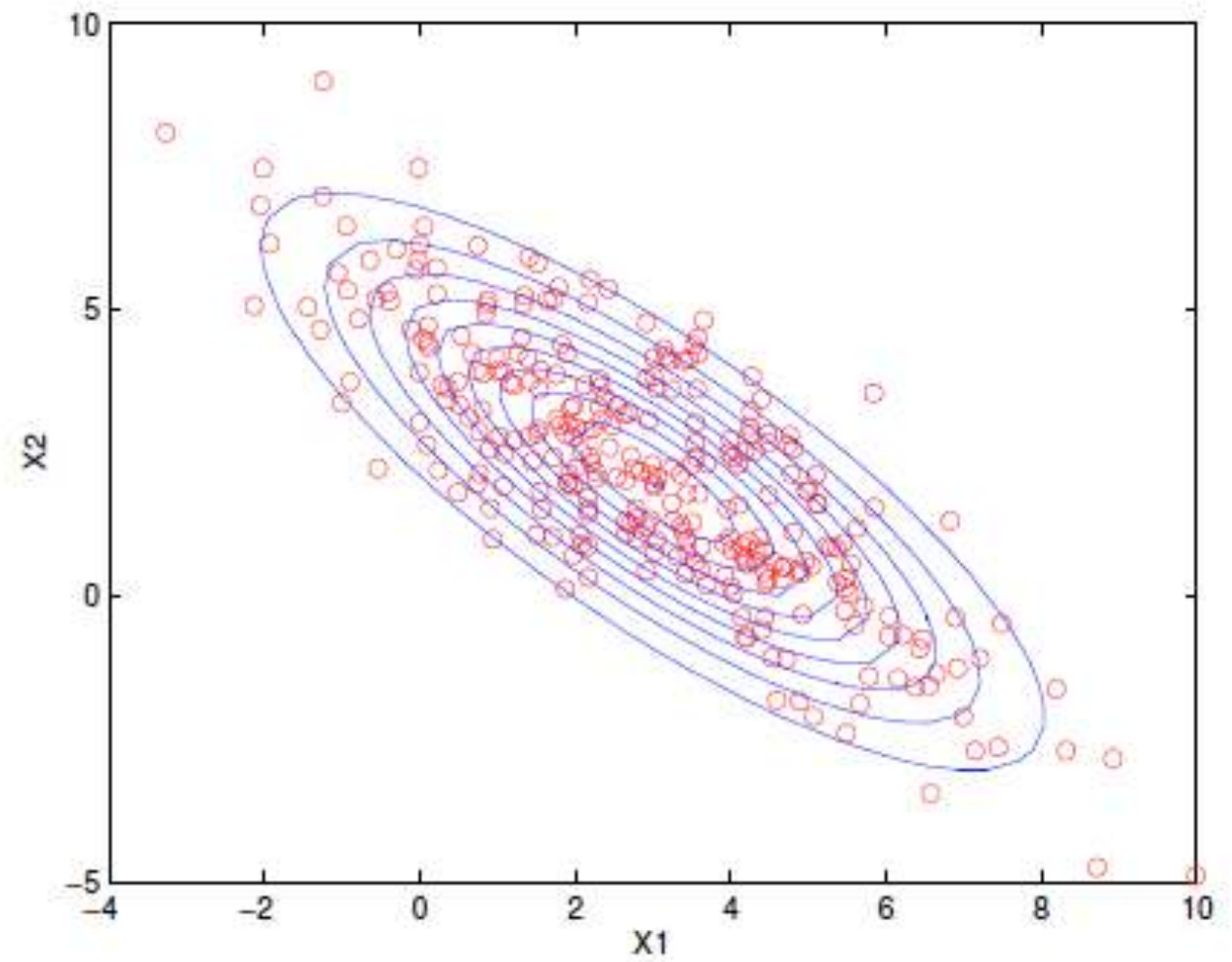
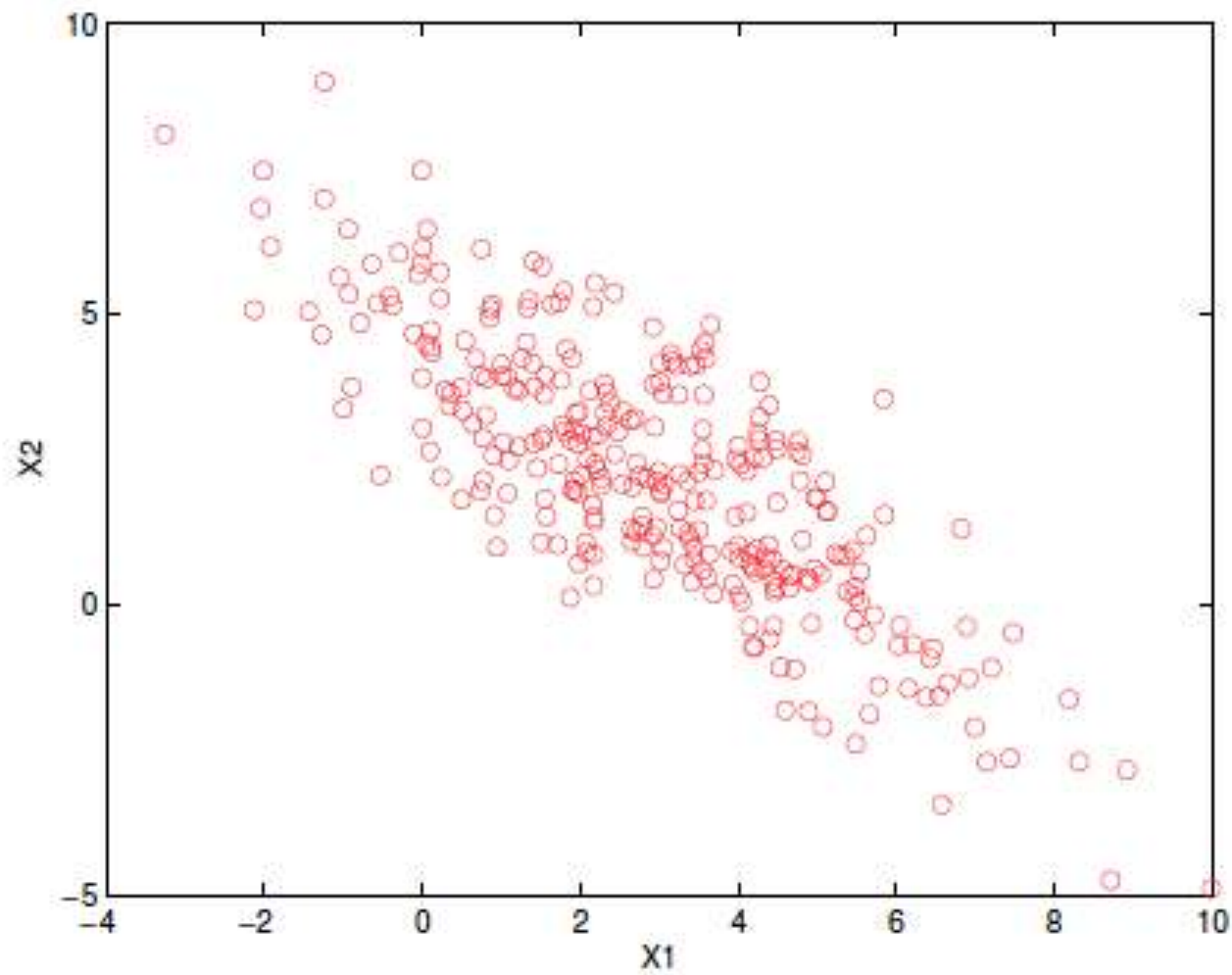
Diagonal Gaussian with different variance

Gaussian Distribution



Full covariance Gaussian distribution

Gaussian Distribution



Finding the parameters of the Model

- ❖ The Gaussian model has the following parameters

$$\theta = (\mu, \Sigma)$$

- ❖ Total number of parameters to be learned for D dimensional data is $D^2 + D$
- ❖ Given N data points $\{\mathbf{x}_i\}_{i=1}^N$ how do we estimate the parameters of model.
 - ❖ Several criteria can be used
 - ❖ The most popular method is the maximum likelihood estimation (MLE).

MLE

Define the likelihood function as $L(\boldsymbol{\theta}) = \prod_{i=1}^N p(\mathbf{x}_i | \boldsymbol{\theta})$

The maximum likelihood estimator (MLE) is

$$\boldsymbol{\theta}^* = \arg \max_{\boldsymbol{\theta}} L(\boldsymbol{\theta})$$

The MLE satisfies nice properties like

- Consistency (convergence to true value)
- Efficiency (has the least Mean squared error).

MLE

For the Gaussian distribution

$$p(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{\sqrt{(2\pi)^D |\boldsymbol{\Sigma}|}} \exp \left\{ -\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^* \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right\}$$

$$L(\boldsymbol{\theta}) = \prod_{i=1}^N p(\mathbf{x}_i|\boldsymbol{\theta})$$

$$\log L(\boldsymbol{\theta}) = -\frac{ND}{2} - \frac{N}{2} \log |\boldsymbol{\Sigma}| - \frac{1}{2} \sum_{i=1}^N \left((\mathbf{x}_i - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x}_i - \boldsymbol{\mu}) \right)$$

To estimate the parameters $\frac{\partial \log L}{\partial \boldsymbol{\mu}} = 0$